

How Much History Is Enough?

Certifying Decision-Relevant Memory in Offline Reinforcement Learning

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Abstract

Partial observability is often handled by conditioning a policy on a fixed window of recent observations and actions. Longer windows can reveal hidden state, but they also enlarge the effective state space and may retain history that improves prediction without changing good decisions. We study how to select the shortest decision-sufficient history window from offline data. In a finite-horizon controlled process, candidate policies use suffixes of the full observable history; importantly, these suffixes need not be Markov. We introduce a cellwise decision ambiguity: the minimum worst-case optimal advantage loss incurred by applying one action distribution to full histories sharing a suffix. Zero ambiguity exactly characterizes the existence of an optimal suffix policy, while cumulative ambiguity upper-bounds its uniform value loss. We then develop Decision-Relevant Memory Selection (DRMS), which converts simultaneous confidence intervals for the full-history optimal Q -function into upper advantage costs and solves small cellwise linear programs. With high probability, DRMS returns the shortest window whose policy loss is certified below a chosen tolerance, without assuming that the selected window is Markov. We prove recovery under a separation condition and matching upper and lower rates in a canonical family: memory identification requires $\Omega(\log(1/\delta)/(p\gamma^2))$ episodes when a decision conflict occurs with probability p and action gap γ . Exact tabular experiments separate next-observation prediction from decision-relevant memory and recover the predicted coverage-gap scaling using only lightweight CPU computation.

Introduction

An agent facing partial observability must decide what part of the past to retain. A common practical answer is frame stacking: concatenate a fixed number of recent observations and actions, then apply an ordinary reinforcement learning algorithm. Choosing this window is consequential. Too little history aliases situations requiring different actions; too much history inflates the effective state space, worsens offline coverage, and invites spurious dependence on nuisance variables. Recurrent policies avoid a fixed window but introduce a larger architecture and tuning problem, and they do not explain how much historical information the task actually requires.

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A tempting solution is to select memory through prediction: retain history as long as it improves prediction of the next observation or reward. This tests whether a truncated observation process is Markov, but the Markov property is stronger than what control requires. Two failures are possible. First, an old nuisance variable may remain highly predictive of future observations even though it never changes the optimal action. Prediction then over-prescribes memory. Second, an old cue may determine which action receives reward while having no effect on subsequent observations. A next-observation test then under-prescribes memory. Figure 1 gives exact instances of both failures.

This paper asks a more direct question:

What is the shortest history window for which offline data can certify a near-optimal policy?

We treat complete observable history as a finite-horizon MDP and compare nested policy classes that use only an m -step suffix. We never assume that the suffix process is Markov. Instead, we inspect whether histories merged by the suffix can share an action without losing value. This leads to a population quantity defined from the full-history optimal advantages. It is zero exactly when a uniformly optimal m -memory policy exists, and it gives an explicit approximation-error bound when exact sufficiency fails.

The population characterization becomes an offline certificate by replacing unknown advantages with simultaneous upper confidence bounds. Each suffix cell requires only a small minimax linear program. The resulting procedure, DRMS, either returns a short policy with a high-confidence value-loss bound or abstains when the data do not support one. Since the guarantee is simultaneous across memory lengths, the same data can be used to choose the window without invalid post-selection reasoning.

Our main contributions are:

- We define *decision ambiguity* for temporal suffix classes and prove monotonicity, an exact decision-sufficiency characterization, and a value-loss sandwich.
- We show that next-observation predictive diagnostics and decision-relevant memory are incomparable: either can demand arbitrarily longer history than the other.
- We introduce DRMS, an interval-based offline certifier that remains valid even though the selected short history is non-Markov.

- We prove exact-memory recovery under an advantage separation and matching upper and lower rates of order $1/(p\gamma^2)$ in a canonical rare-context family.
- We provide transparent tabular code and theorem-driven experiments. Across 12 coverage–gap settings, the empirical episodes needed for 50% recovery scale with $1/(p\gamma^2)$ with log–log slope 0.94.

Related Work

Memory and partial observability. General POMDP planning can require a belief over latent states (Kaelbling, Littman, and Cassandra 1998). Known-model analyses bound finite-window approximation under nonlinear-filter stability (Kara and Yüksel 2022), while short-memory POMDP theory assumes that a recent window decodes latent state and obtains complexity depending on the window length rather than the full horizon (Efroni et al. 2022). Recent metric belief approximation gives policy-conditional finite-memory loss bounds (Kim 2026). Offline POMDP learning has been studied through finite-memory Bellman operators and structural assumptions such as undercompleteness (Guo et al. 2022), and offline safe policy improvement with finite-state controllers assumes a fixed memory structure (Simão, Suilen, and Jansen 2023). Memory traces provide a compact alternative to raw windows and yield offline on-policy evaluation guarantees (Eberhard, Muehlebach, and Vernade 2025); recurrent model-free policies can also be strong empirical POMDP baselines (Ni, Eysenbach, and Salakhutdinov 2022). These works ask how to learn or improve a policy with a specified memory mechanism. We instead select among nested raw-history suffixes from fixed data and certify decision loss without requiring the selected suffix to be Markov.

Prediction and state representation. Predictive-state ideas represent history through forecasts of future events (Littman, Sutton, and Singh 2001). Recent Markov Violation Scores diagnose non-Markov observations by testing whether history reduces prediction error (Mysore 2026). Our counterexamples show why next-observation predictive violation is not itself a policy-memory criterion. The broader state-abstraction literature characterizes value- and policy-preserving mappings (Li, Walsh, and Littman 2006; Abel, Hershkowitz, and Littman 2016; Abel et al. 2020), and action-sufficient representation learning targets variables needed for control (Huang et al. 2022). We specialize decision sufficiency to temporal suffix partitions and add offline selection, confidence certification, and coverage-dependent lower bounds.

Offline RL and model selection. Offline RL uses pessimism to avoid unsupported decisions (Jin, Yang, and Wang 2021; Shi et al. 2022). Related work connects pessimistic offline RL with imitation learning (Rashidinejad et al. 2021). ModBE selects among nested value-function classes using oracle inequalities (Lee et al. 2022). Our nested objects are policy information sets rather than Bellman model classes: a shorter suffix may be non-Markov and Bellman incomplete yet still contain an optimal policy. We use full-history confidence intervals only to certify the action compromises induced by each suffix partition.

Setting and Objective

Consider a finite-horizon controlled process with stages $h \in [H]$, bounded rewards $R_h \in [0, 1]$, and finite action sets $\mathcal{A}_h$. Let $X_h \in \mathcal{X}_h$ be the complete observable action–observation history before action A_h . The complete history is a valid Markov state, with reward mean $r_h(x, a)$ and transition kernel $P_h(\cdot \mid x, a)$. Let Q_h^*, V_h^* be the corresponding optimal action-value and value functions.

For memory length m , the map $\phi_{m,h}(x)$ retains the latest m observations and intervening actions, or all available history when $h < m$. A policy belongs to

$$\Pi_m = \{\pi : \pi_h(\cdot \mid x) = \pi_h(\cdot \mid \phi_{m,h}(x))\}.$$

The classes are nested. We emphasize that $Z_h = \phi_{m,h}(X_h)$ need not be Markov for $m < H$.

We seek the shortest m supporting small uniform loss. Define

$$\mathcal{E}_m = \inf_{\pi \in \Pi_m} \max_{h,x} [V_h^*(x) - V_h^\pi(x)].$$

We use uniform loss for a clean distribution-free theory and defer occupancy weighting to the limitations.

Decision-Relevant Memory

Define the optimal advantage loss

$$A_h^*(x, a) = V_h^*(x) - Q_h^*(x, a) \geq 0.$$

For suffix z , let $C_{m,h}(z) = \{x : \phi_{m,h}(x) = z\}$ be the full histories that an m -memory policy must treat identically.

Definition 1 (Decision ambiguity). *The ambiguity of one suffix cell is*

$$\eta_{m,h}(z) = \min_{p \in \Delta(\mathcal{A}_h)} \max_{x \in C_{m,h}(z)} \mathbb{E}_{a \sim p} A_h^*(x, a).$$

Let $\eta_{m,h} = \max_z \eta_{m,h}(z)$ and $G_m = \sum_{h=1}^H \eta_{m,h}$.

The inner policy is allowed to randomize. For deterministic policies, zero ambiguity means that every suffix cell has a common optimal action. The quantity is computable by a linear program with $|\mathcal{A}_h| + 1$ variables per cell.

Theorem 1 (Population characterization). *For every m :*

1. $G_{m+1} \leq G_m$.
2. $G_m = 0$ if and only if there exists $\pi \in \Pi_m$ with $V_h^\pi(x) = V_h^*(x)$ for every h, x .
3. The best attainable uniform loss satisfies

$$\max_h \eta_{m,h} \leq \mathcal{E}_m \leq G_m.$$

The cellwise minimax policy π^m additionally obeys

$$V_h^*(x) - V_h^{\pi^m}(x) \leq \sum_{t=h}^H \eta_{m,t}.$$

The proof uses no Markov property for the suffix. For the upper bound, let $D_h(x) = V_h^*(x) - V_h^{\pi^m}(x)$. Adding and subtracting the full-history optimal continuation gives

$$\begin{aligned} D_h(x) &= \mathbb{E}_{a \sim \pi_h^m} A_h^*(x, a) \\ &\quad + \mathbb{E}_{a, X'} D_{h+1}(X') \\ &\leq \eta_{m,h} + \|D_{h+1}\|_\infty. \end{aligned}$$

Backward induction proves the claim. Conversely, any suffix policy must use one action distribution throughout a cell, while replacing its continuation by V^* can only increase value; this yields the lower bound. Full proofs are in the supplement.

For a behavior policy μ , write $Z_h^m = \phi_{m,h}(X_h)$ and define the oracle next-observation residual

$$R_m^{\text{obs}} = \sum_h H_\mu(O_{h+1} \mid Z_h^m, A_h) - \sum_h H_\mu(O_{h+1} \mid X_h, A_h).$$

This mirrors observation-prediction diagnostics, not the full controlled Markov condition. If the conditional joint law of reward and next suffix is identical within every suffix cell for every action, then an optimal suffix policy exists and $G_m = 0$.

Proposition 1 (Observation-prediction separation). *For any horizon H , there are controlled processes under a uniform behavior policy in which: (i) $R_m^{\text{obs}} = 0$ for every m , but $G_m = \gamma/2$ for all $m < H$ and $G_H = 0$; and (ii) $G_m = 0$ for every m , but $R_m^{\text{obs}} = 1$ bit for all $m < H$. Thus an observation-prediction diagnostic can require arbitrarily more or less history than decision sufficiency.*

Offline Certification with DRMS

Assume an offline procedure produces simultaneous intervals

$$L_h(x, a) \leq Q_h^*(x, a) \leq U_h(x, a) \quad \text{for all } h, x, a \quad (1)$$

with probability at least $1 - \delta$. Our core result is modular: any valid full-history Q^* intervals may be used.

A naive upper advantage, $\max_b U_h(x, b) - L_h(x, a)$, is unnecessarily loose because comparing action a with itself has exactly zero loss. We instead define

$$\bar{A}_h(x, a) = \max \left\{ 0, \max_{b \neq a} [U_h(x, b) - L_h(x, a)] \right\}, \quad (2)$$

with $\bar{A}_h = 0$ at one-action stages. On event (1), $A_h^*(x, a) \leq \bar{A}_h(x, a)$.

Replace A_h^* by $\bar{A}_h$ in Definition 1 to obtain robust ambiguities $\bar{\eta}_{m,h}(z)$ and $\bar{G}_m = \sum_h \max_z \bar{\eta}_{m,h}(z)$, along with the robust cellwise policy $\bar{\pi}^m$.

Decision-Relevant Memory Selection. Given tolerance ϵ , DRMS performs:

1. estimate simultaneous full-history intervals (L, U) ;
2. for each m, h, z , solve the minimax LP using costs $\bar{A}_h$;
3. return the smallest m with $\bar{G}_m \leq \epsilon$ and its policy $\bar{\pi}^m$;
4. if no candidate passes, return the longest candidate with an explicit abstention flag.

Theorem 2 (High-confidence value certificate). *On event (1), simultaneously for every memory length m ,*

$$V_h^*(x) - V_h^{\bar{\pi}^m}(x) \leq \sum_{t=h}^H \bar{\eta}_{m,t} \quad \text{for all } h, x.$$

Consequently, whenever DRMS certifies $\bar{G}_m \leq \epsilon$, its selected policy has uniform value loss at most ϵ .

The theorem follows by the same backward recursion as Theorem 1, now using $A_h^* \leq \bar{A}_h$. Since the intervals are simultaneous, the result holds for the data-selected memory without a second holdout set.

Proposition 2 (Tightness and recovery). *Suppose every interval has one-sided error at most w . Then*

$$G_m \leq \bar{G}_m \leq G_m + 2Hw.$$

If $m_{\text{dec}} = \min\{m : G_m = 0\} > 1$, let $\Gamma = \min_{m < m_{\text{dec}}} G_m$. If $2Hw < \Gamma$, every threshold $\epsilon \in [2Hw, \Gamma)$ makes DRMS recover m_{dec} .

This yields a certificate-complexity tradeoff: longer memory weakly decreases ambiguity, while DRMS returns the shortest window meeting a user-specified loss budget. Statistical uncertainty is inherited from the common full-history confidence set. The present method therefore certifies memory parsimony rather than balancing separately estimated models.

Tabular Instantiation and Statistical Limit

For transparency, we instantiate Eq. (1) in a finite full-history MDP from n independent behavior-policy episodes. For each (h, x, a) , empirical rewards use a union-bounded Hoeffding radius, and empirical transitions use an L_1 radius based on Weissman et al. (2003). Optimistic and pessimistic dynamic programming propagate these model intervals using

$$|(P - \hat{P})^\top f| \leq \frac{1}{2} \|P - \hat{P}\|_1 \text{span}(f).$$

A backward induction gives simultaneous Q^* coverage. Unobserved pairs receive the full reward radius and transition radius two; therefore the method abstains rather than extrapolating through unsupported histories.

Coverage is not merely an artifact of our algorithm. The following lower bound identifies the intrinsic difficulty of deciding whether an old cue is needed.

Theorem 3 (Memory-order lower bound). *Fix $0 < p < 1$, $0 < \gamma \leq 1/2$, and $0 < \delta < 1/4$. There are two horizon-two processes observed under the same uniform behavior policy such that their shortest decision-sufficient memories are one and two, respectively, and any estimator that identifies the correct memory with error at most δ on both requires*

$$n \geq \frac{\log(1/(4\delta))}{6p\gamma^2}.$$

In the construction, a cue is rare with probability p . In the first model, action zero is better by γ after either cue. In the second, the optimal action flips only after the rare cue. The one-episode KL divergence is at most $6p\gamma^2$, and the result follows from the Bretagnolle–Huber inequality (Tsybakov 2009). The theorem explains why memory selection is hard precisely when decision-conflicting histories are rare or their action gaps are small.

Corollary 1 (Matching upper rate in the two-point family). *Let $p \leq 1/2$, and let M be the number of full-history state-action pairs. Tabular DRMS with tolerance $\gamma/4$ and confidence level $\delta/2$ recovers the correct memory in both models with probability at least $1 - \delta$ whenever*

$$n \geq \frac{32 \log(8M/\delta)}{p\gamma^2}.$$

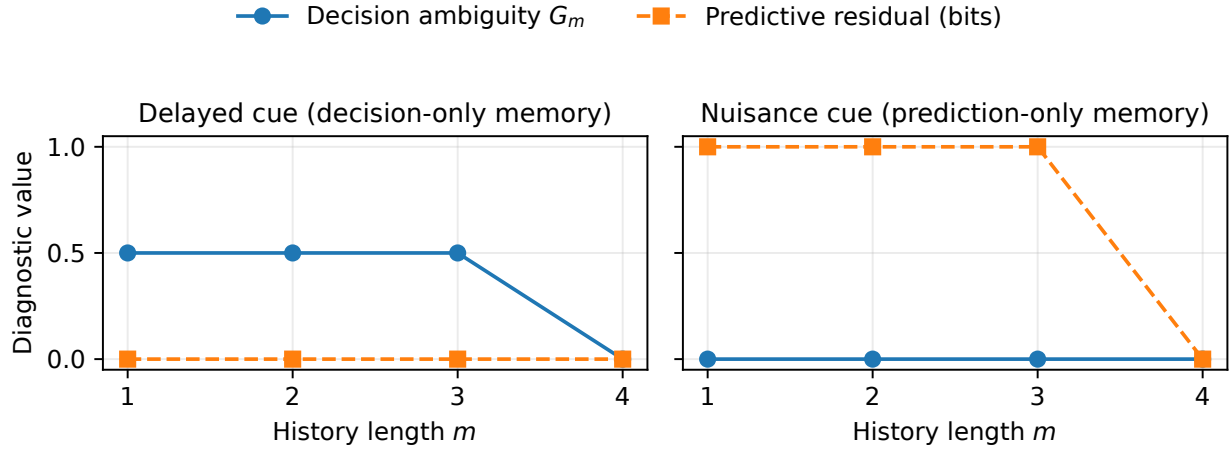


Figure 1: **Next-observation prediction and decision-relevant memory can disagree in both directions.** In the delayed-cue task (left), a stage-one cue determines the final rewarding action, but future observations are constant: $R_m^{\text{obs}} = 0$, whereas $G_m = 1/2$ until the cue enters the suffix. In the nuisance task (right), the cue predicts the terminal observation but never changes the optimal action, so $G_m = 0$ at every memory length. Values are exact for $H = 4$ and a balanced binary cue.

The proof combines a binomial lower tail for every final state–action count with reward confidence radii small enough to separate the two actions. Thus the lower bound is tight in p and γ , up to logarithms and constants.

Experiments

Our experiments are deliberately small: their role is to validate theorem predictions, not to claim deep-control state of the art. All values in the population study are computed exactly. Finite-sample runs use only tabular counts, dynamic programming, and two-action cellwise minimax solutions.

Prediction can select the wrong memory in both directions.

Figure 1 uses horizon four. In the delayed-cue process, the cue leaves the suffix until $m = 4$; a memoryless final action incurs minimax loss $1/2$, while the next observation is always deterministic. In the nuisance process, the cue contributes one bit of terminal predictive information but action zero is optimal for both cues. The computed policy loss equals G_m in both examples.

Tolerance adapts to decision relevance. We vary the delayed-cue reward gap γ at horizon five. Every insufficient suffix has population ambiguity $\gamma/2$, so tolerance ϵ should select memory one exactly when $\gamma \leq 2\epsilon$, and full memory otherwise. The implementation recovers these four phase transitions exactly; the corresponding plot is in the supplement.

Coverage–gap scaling and calibration. Figure 2 evaluates the matching dependence in Theorem 3 and Corollary 1. We run 2,880 independent datasets over 12 (p, γ) combinations. The empirical 50%-recovery threshold has a log–log slope of 0.935 against $1/(p\gamma^2)$. In a separate calibration sweep with 1,600 datasets, four values of p , and fixed $\gamma = 0.5$, $\epsilon = 0.1$, simultaneous Q -interval coverage and policy certificate validity are both 100%; no run falsely certifies a shorter

memory. These observations are sanity checks rather than substitutes for the finite-sample theorem, but they show that the conservative construction becomes informative at the predicted effective sample scale.

Reproducibility. The entire experiment suite runs on CPU. The repository includes exact MDP constructors, interval dynamic programming, cellwise solvers, raw and summary CSVs, figure scripts, and unit tests. The main scaling experiment uses 30 seeds; all configuration values are encoded in the scripts.

Limitations and Conclusion

The present theory enumerates a known finite maximum history and uses uniform loss over all feasible full contexts. This is intentionally conservative: long histories can be exponentially numerous, and a rare but irrelevant context can block certification. The next theoretical step is an occupancy-weighted ambiguity with a concentrability-based value guarantee. Function approximation, continuous observations, and jointly learning a representation and its memory length remain open. Our tabular confidence set is also designed for transparency rather than sharp constants.

Within this scope, the results establish a simple principle: memory should be selected according to the action regret created by forgetting, not according to prediction alone. Decision ambiguity makes that principle exact at the population level, while DRMS turns it into an offline, high-confidence certificate that can abstain when coverage is inadequate. The matching-form coverage–gap lower bound clarifies when such certification is statistically possible.

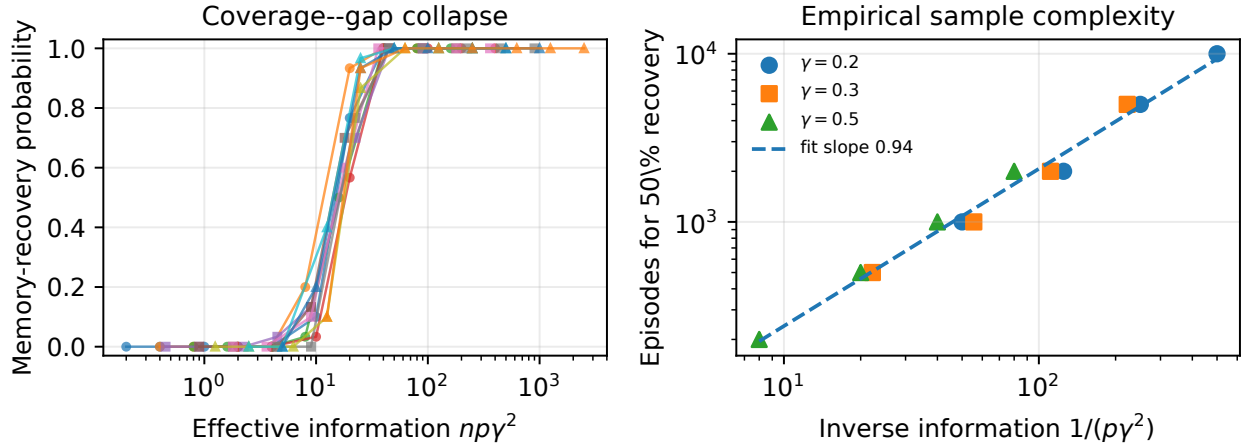


Figure 2: **Finite-sample behavior follows the information-theoretic scaling.** We vary rare-cue probability $p \in \{0.05, 0.1, 0.2, 0.5\}$, action gap $\gamma \in \{0.2, 0.3, 0.5\}$, and eight dataset sizes, with 30 seeds per point. The tolerance is $\gamma/4$. Left: all 12 recovery curves nearly collapse against $np\gamma^2$. Right: the first grid size attaining at least 50% correct certification is approximately linear in $1/(p\gamma^2)$; the fitted log–log slope is 0.94.

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